

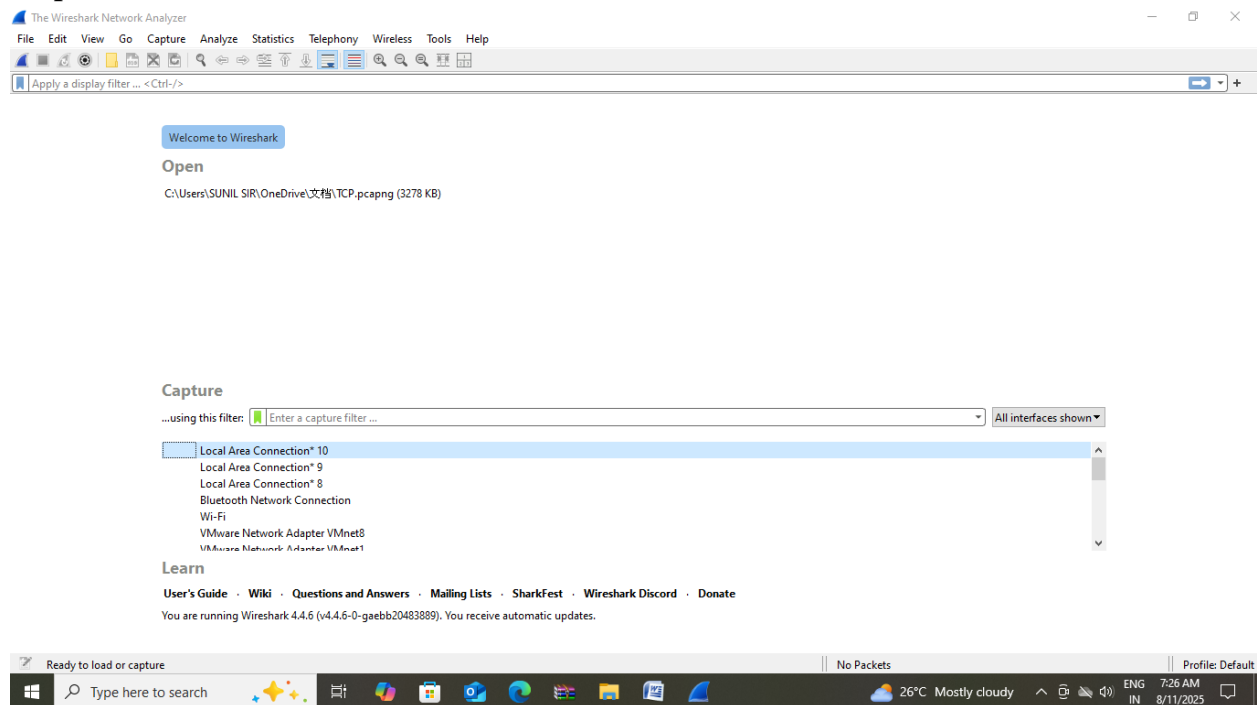
Task 5: Capture and Analyze Network Traffic Using Wireshark.

- **Objective:** Capture live network packets and identify basic protocols and traffic types.
- **Tools:** Wireshark (free).
- **Deliverables:** A packet capture (.pcap) file and a short report of protocols identified.

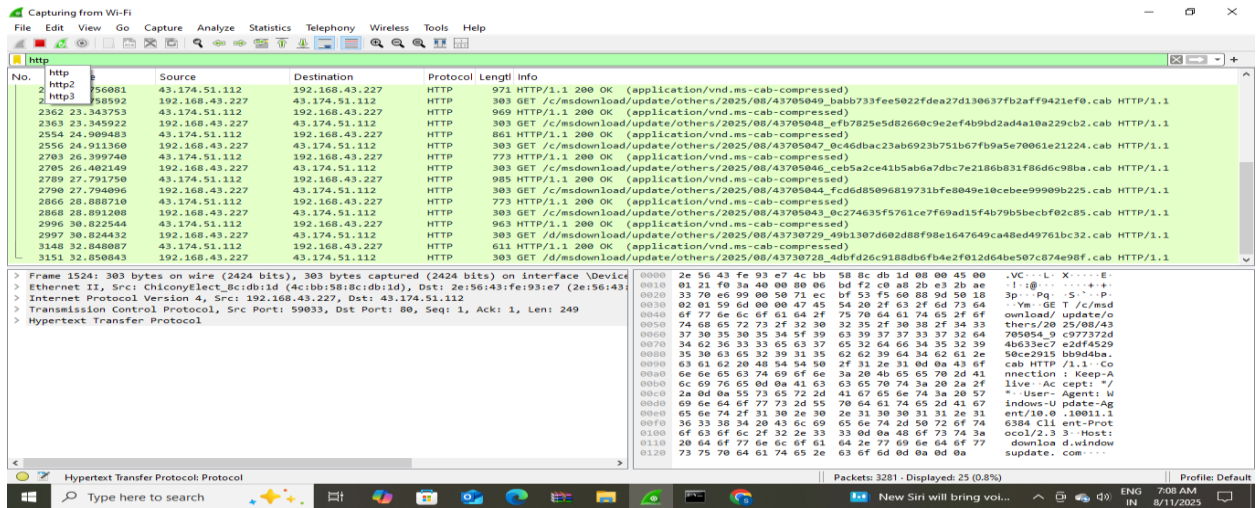
Hints/Mini Guide:

1. Install Wireshark.
2. Start capturing on your active network interface.
3. Browse a website or ping a server to generate traffic.
4. Stop capture after a minute.
5. Filter captured packets by protocol (e.g., HTTP, DNS, TCP).
6. Identify at least 3 different protocols in the capture.
7. Export the capture as a .pcap file.
8. Summarize your findings and packet details.

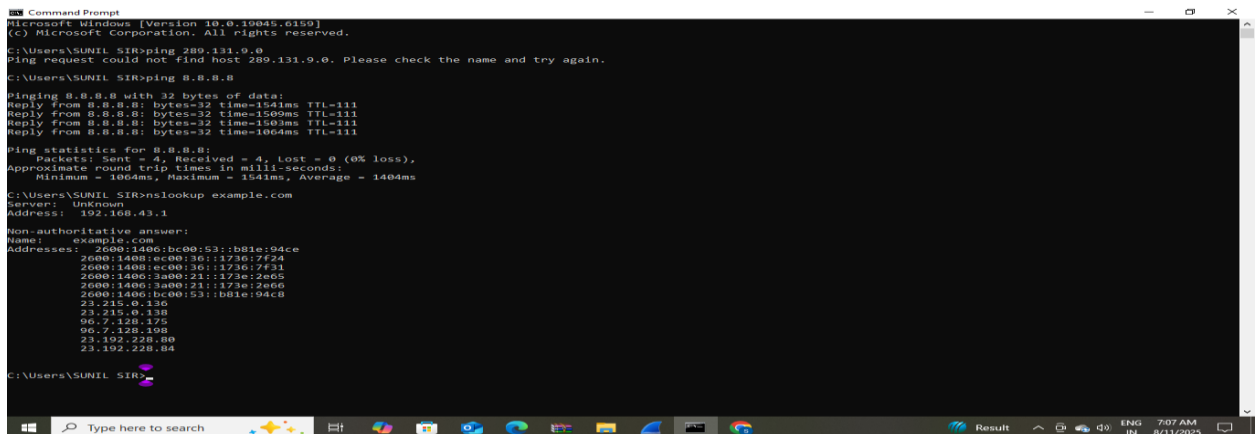
Step1. Install Wireshark



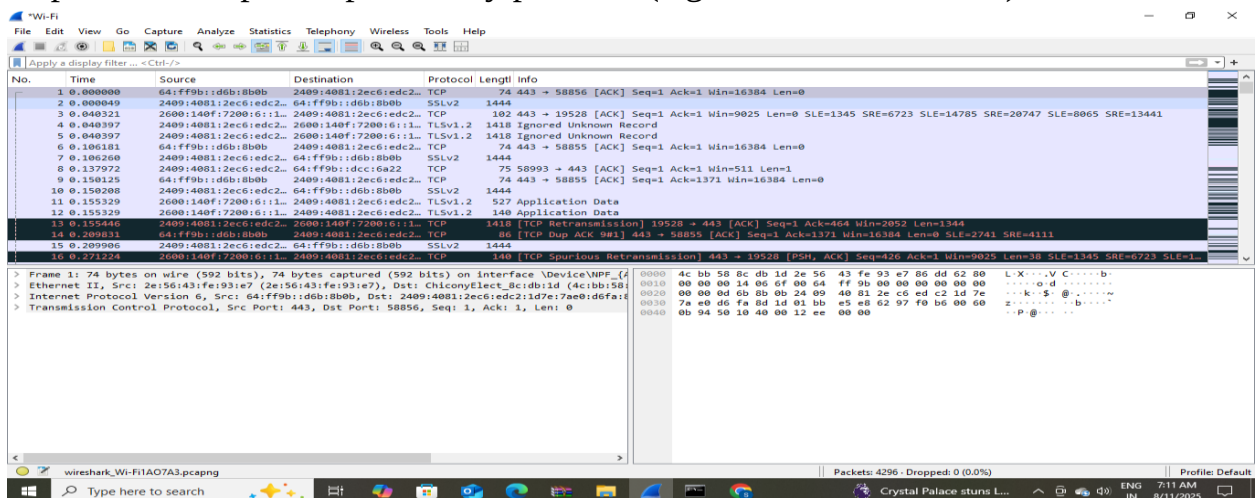
Step 2. Start capturing on your active network interface.



Step 3: Browse a website or ping a server to generate traffic.



Step 4: Filter captured packets by protocol (e.g., HTTP, DNS, TCP).



Step 6:. Identify at least 3 different protocols in the capture.

The screenshot shows a Wireshark capture of network traffic. The packet list on the left shows a series of TCP packets (No. 80-94) and SSL packets (No. 95-100). The packet details pane on the right shows the selected packet (No. 80) with the following information:

- Destination: 2409:4081:2ec6:edc2::
- Protocol: TCP
- Length: 86
- Info: [TCP Dup ACK 169#1] 443 → 58955 [ACK] Seq=1 Ack=9591 Win=16384 Len=0 SLE=10961 SRE=12331

The packet bytes pane on the right shows the raw data of the selected packet, which is a TCP segment with a sequence number of 1 and an acknowledgment number of 9591.

The screenshot shows a Wireshark capture of network traffic. The packet list on the left shows a series of HTTP packets (No. 80-94) and SSL packets (No. 95-100). The packet details pane on the right shows the selected packet (No. 80) with the following information:

- Source: 64:ff9b:d6b:8b0b
- Destination: 2409:4081:2ec6:edc2::
- Protocol: TCP
- Length: 110
- Info: 443 → 58955 [ACK] Seq=1 Ack=107614 Win=16384 Len=0 SLE=106244 SRE=106840 SLE=124954 SRE=143234 SLE=118574

The packet bytes pane on the right shows the raw data of the selected packet, which is a TCP segment with a sequence number of 1 and an acknowledgment number of 107614.